complaint-router

Lab 2: data pipeline, versioning, baseline and MLflow

KSE MLOps · Lab 2 · Zakhar Staryk · github.com/goodfrog33/kse-mlops-complaint-router · tag lab-2

39
monthly batches through Airflow

0.638 vs 0.384
macro-F1, model vs rule

0.0000
reproduce difference

At a glance

Data versioned

- 117,000 complaints, 39 months, DVC
- MinIO remote + public read-only mirror

Pipeline

- Airflow 3.3.1, six tasks, one month per run
- Reruns leave identical rows

Quality gate

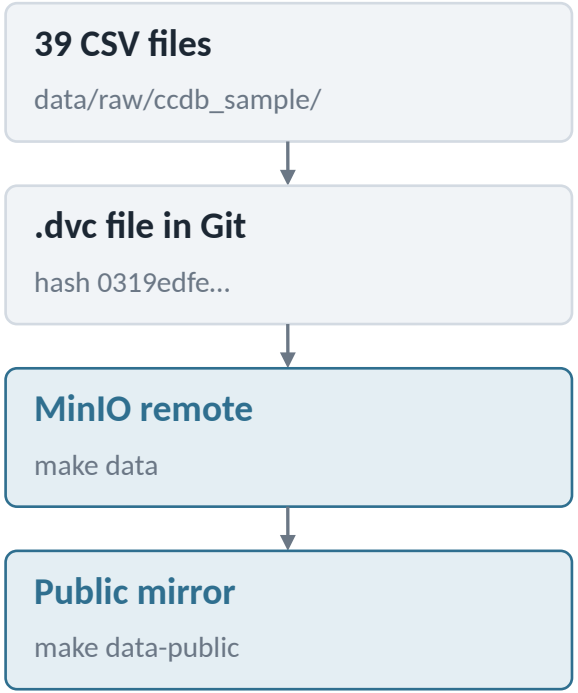
- Fails on purpose (fault injection)
- Fails for real: 2025-12 → 2026-06

Model and lineage

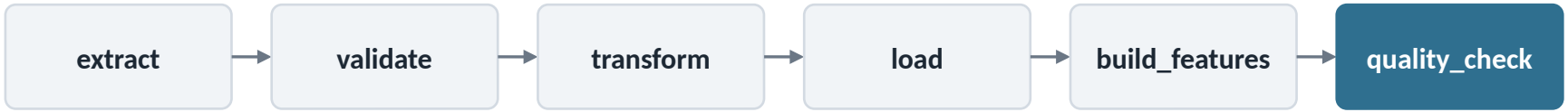
- TF-IDF 0.638 vs keyword rule 0.384
- MLflow v2 @champion, reproduced exactly

Raw data is versioned with DVC

Sample	39 monthly CSVs × 3,000 complaints (66 MB)
Drawn by	hash of complaint_id: uniform, byte-identical reruns
Data version	`0319edfe2ad83a39d4d8241602a462f8`
Remotes	MinIO s3://dvc · public HTTP mirror
Pull	make data · make data-public
Checksum	`0bba9005abf7d408af110f7a192fc13a`



One Airflow run per month, never the clock



Batch

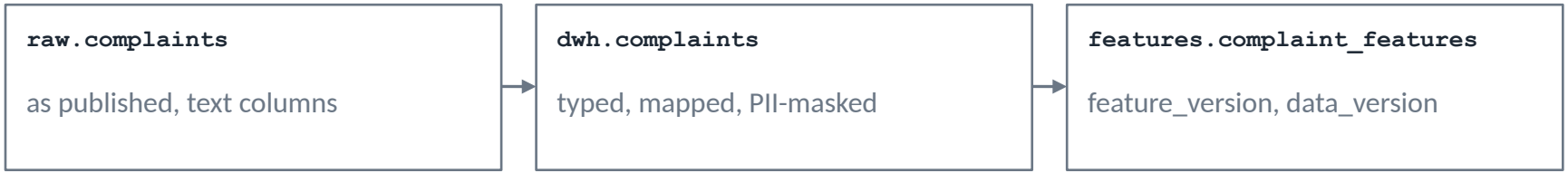
`data_interval_start`

Schedule (@monthly)

`CronDataIntervalTimetable`

Concurrency

6 months at once: 39 in 113 s



PostgreSQL warehouse

Rerunning a month leaves identical rows

Run 1 · 2024-06	rows
raw.complaints	3000
dwh.complaints	3000
features	1013
checks passed	5 / 5

Run 2 · 2024-06	rows
raw.complaints	3000
dwh.complaints	3000
features	1013
checks passed	5 / 5

Content hash, before and after

e15aa30f…

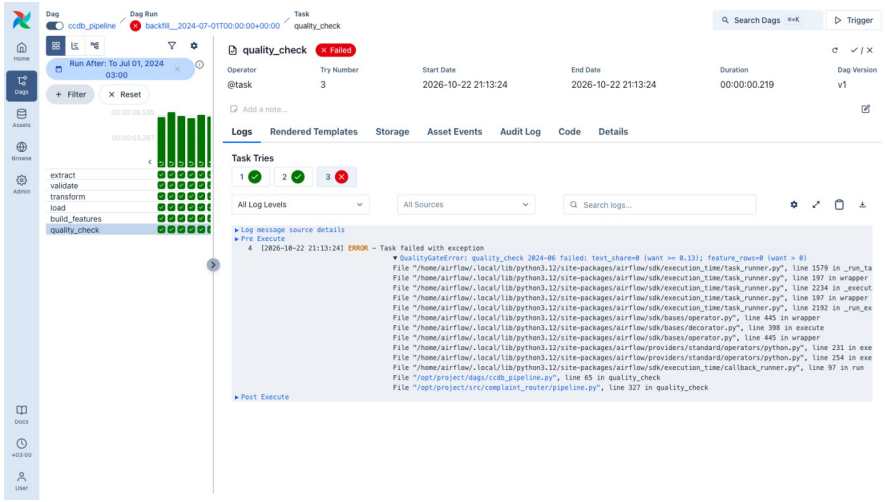
IDENTICAL

md5 of every feature row of the month

Why: every load deletes the month, then inserts it, in one transaction

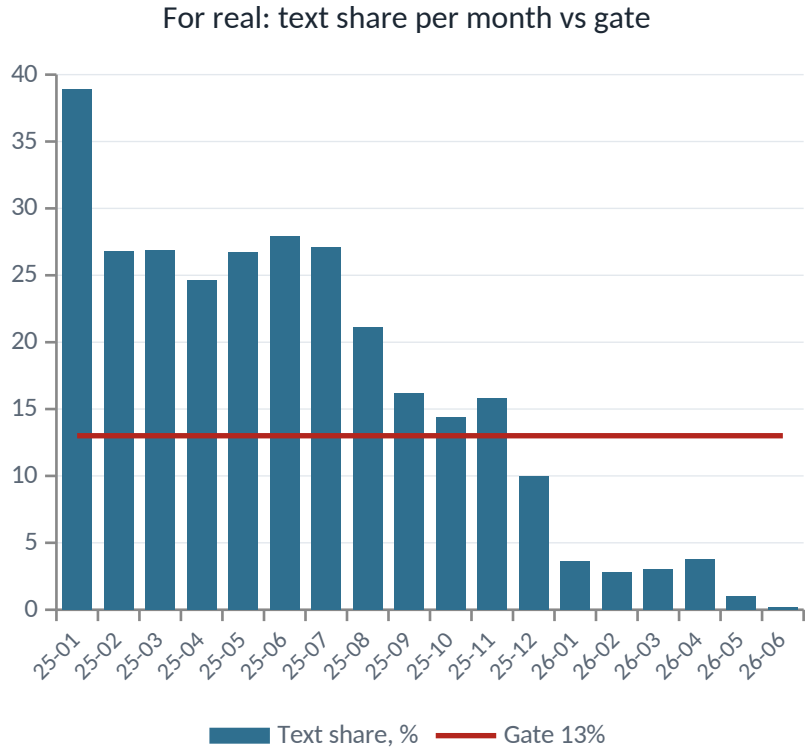
```
delete from <table> where batch_month = %s;  copy <table> from stdin
```

The quality gate fails when it should



On purpose: FAULT=null_narratives

```
text_share=0 (want >= 0.13);
feature_rows=0 (want > 0)
```



Every feature has a serving-time source

Feature	Training	Serving
text	dwh narrative, clean_text_expr	request, clean_text (tested equal)
submitted_via	CFPB field	request; missing → unknown
state	CFPB field	request; missing → unknown
company	normalized name	dropped: sources differ

Feast: point-in-time join

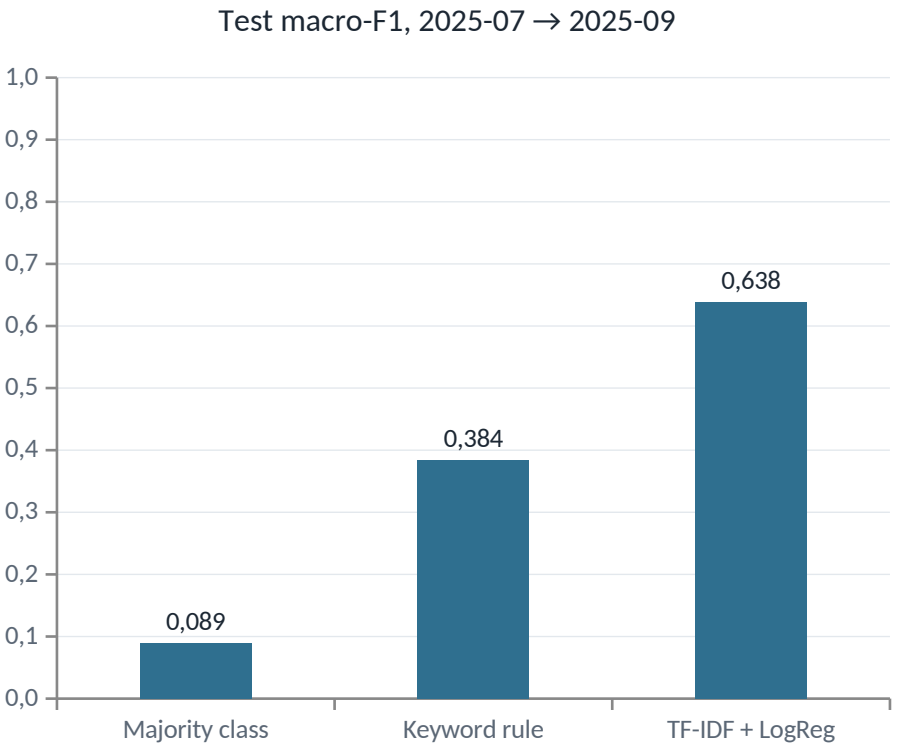
Complaint on 2024-07-15; feature rows at 07-01 and 08-01

Company	Got	Later
Ally Financial	3	5
American Express	6	8
Affirm	4	3

EARLIER ROW EVERY TIME

```
make feast-demo
```

The model beats the rule on every split

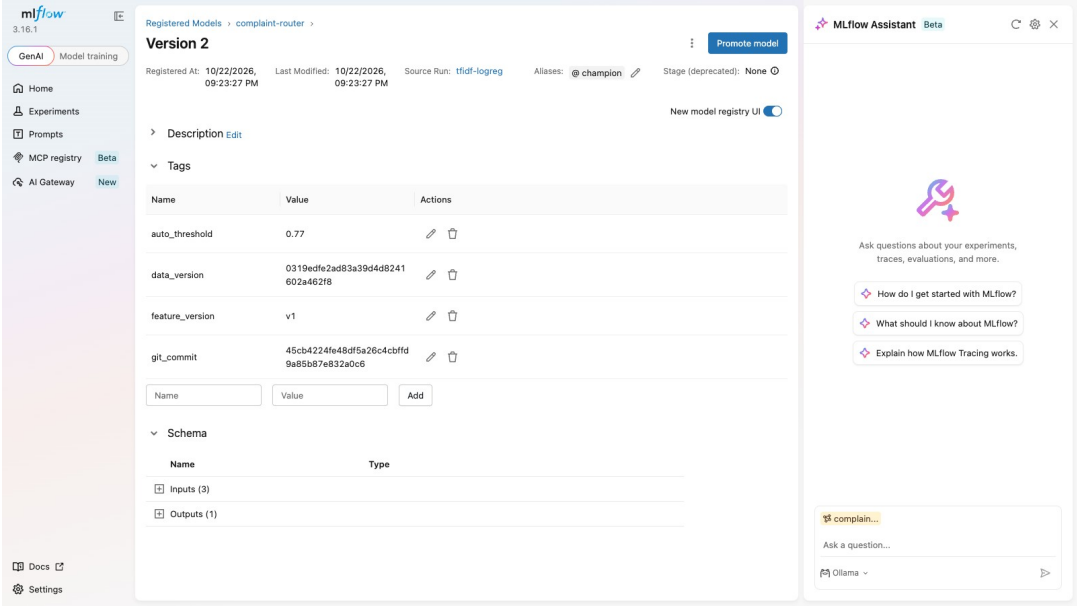


Backtest month	Model	Rule
2025-07	0.652	0.374
2025-08	0.682	0.377
2025-09	0.716	0.405
mean ± std	**0.684 ± 0.026**	0.385 ± 0.014

Random split: 0.591

No inflation on this uniform 0.5% sample; time split kept because it matches use

Every run carries its data version and commit



Server

MLflow 3.16.1 · Postgres backend · MinIO artifacts

Registered

complaint-router v2 · @champion

Lineage tags

data_version · git_commit

Rule baselines, backtest folds and the random-split check are logged with the same tags.

One command reproduces the champion

1. Read tags from registry → 2. git worktree at commit → 3. dvc pull + hash check → 4. Rebuild features, retrain

```
$ make reproduce
version 2: commit 45cb4224fe, data 0319edfe…, test_macro_f1 0.6378
registered 0.6378  reproduced 0.6378  |diff| 0.0000  tolerance 0.005  -> OK
```

0.0000

difference on the same platform

0.005

tolerance: BLAS and thread differences

Lab 2 rows of the maturity scorecard

Property	Evidence
Data versioned	make data-public restores 39 files, checksum 0bba9005…
Pipeline idempotent	2024-06 twice: 3000 / 3000 / 1013, same hash
Quality gate present	fault fails quality_check; 2025-12 → 2026-06 fail for real
Experiments tracked	every run: params, metrics, data_version, git_commit
Model registered with lineage	v2 @champion; make reproduce → diff 0.0000

Run it

```
make setup
make up
make data-seed
make data
make backfill
make train
make reproduce
```

CI: lint, unit and pipeline tests